ECEN 662 Estimation & Detection Theory **Name:**
Spring 2026 — Homework #3
Due date: Sunday, February 15

Problem 1 (20 Points): Let $X = (N, Y)$, where N is a random variable taking values from $\mathcal{Z}_{++}$ with a fixed pmf $p(n)$. Having observed $N = n$, mother nature performs n Bernoulli trials with success probability θ, observing a total of Y successes.

i) (10 Points) Show that X itself is a minimal sufficient statistic for θ.

ii) (10 Points) Show that $\hat{\theta}(X) = \frac{Y}{N}$ is an unbiased estimator of θ and

$$\mathrm{Var}_\theta[\hat{\theta}(X)] = \theta(1-\theta)\mathbb{E}\left[\frac{1}{N}\right].$$

Hint: It may be useful to consider the law of total variance.

Problem 2 (15 Points): Let $X \sim f_\theta(x)$, where $\theta \in \Theta$ is the unknown parameter. For any $\theta, \theta' \in \Theta$, let

$$\psi_{\theta',\theta}(x) := \frac{f_{\theta'}(x)}{f_\theta(x)} - 1.$$

i) (5 Points) Show that for any scalar statistic $T(x)$,

$$\mathbb{E}_\theta[\psi_{\theta',\theta}(X)T(X)] = \mathbb{E}_{\theta'}[T(X)] - \mathbb{E}_\theta[T(X)], \quad \forall \theta, \theta' \in \Theta.$$

In particular, we have

$$\mathbb{E}_\theta[\psi_{\theta',\theta}(X)] = 0, \quad \forall \theta, \theta' \in \Theta.$$

ii) (10 Points) Use the Cauchy-Schwarz inequality to show that

$$\mathrm{Var}_\theta[\hat{g}(X)] \geq \sup_{\theta' \in \Theta} \frac{(\phi(\theta') - \phi(\theta))^2}{\mathbb{E}_\theta\left[\psi^2_{\theta',\theta}(X)\right]}, \quad \forall \theta \in \Theta$$

for any estimator $\hat{g}(X)$ of $g(\theta)$, where

$$\phi(\theta) := \mathbb{E}_\theta[\hat{g}(X)].$$

Problem 3 (15 Points): Let $X \sim p_\theta(x)$ and $T = T(X)$ be a statistic for θ. Let $s_\theta(x)$ and $I_X(\theta)$ be the score function and Fisher information of θ with respect to the observation X, and let $\tilde{s}_\theta(t)$ and $I_T(\theta)$ be the score function and Fisher information of θ with respect to the statistic T.

i) (10 Points) Show that

$$\tilde{s}_\theta(T) = \mathbb{E}_\theta[s_\theta(X)|T], \quad \forall \theta.$$

ii) (5 Points) Show that

$$I_T(\theta) \preceq I_X(\theta), \quad \forall\theta,$$

where the equalities hold when T is sufficient statistic of θ. Here, $A \preceq B$ means that $B - A$ is positive definite.

Problem 4 (20 Points): Let $X \sim f_\theta(x) = \frac{1}{\theta} g\left(\frac{x}{\theta}\right)$, where $\theta > 0$ is the unknown parameter and g is a fixed density function.

i) (5 Points) Show that

$$I(\theta) = \frac{1}{\theta^2} \mathbb{E}_g \left[\left(\frac{Y g'(Y)}{g(Y)} + 1 \right)^2 \right]$$

where the expectation is over $Y \sim g(y)$.

ii) (5 Points) Let $\eta = \log\theta$. Compute $I(\eta)$.

iii) (10 Points) Compute $I(\theta)$ when g is $Cauchy(0, 1)$.